MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC ASSESSMENT TEST – 10** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

MATHS MANTHAN | $6 \times 1 = 6$ Marks

Q1. If the product of two consecutive positive integers is 306, then the integers are: [1]

(A) 16, 17 **(B)** 17, 18
(C) 18, 19 **(D)** 15, 16

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 17, 18

Let the consecutive positive integers be x and $x+1$ ($x > 0$).

Formula Used: $x(x+1) = 306 \implies x^2 + x - 306 = 0$

Factorising ($18 \times (-17) = -306$ and $18 - 17 = 1$):

$$x^2 + 18x - 17x - 306 = 0 \implies (x+18)(x-17) = 0 \implies x = 17 \quad (\text{reject } x = -18 < 0)$$

So the integers are 17 and $17 + 1 = 18$. Hence, option **(B)** is correct. **[1 mark]**

Q2. The discriminant of the quadratic equation $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ is: [1]

(A) 32 **(B)** -32
(C) 16 **(D)** 24

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 32

Comparing $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ with $ax^2 + bx + c = 0$, we have $a = \sqrt{3}$, $b = -2\sqrt{2}$, and $c = -2\sqrt{3}$.

Formula Used: Discriminant $D = b^2 - 4ac$

$$D = (-2\sqrt{2})^2 - 4(\sqrt{3})(-2\sqrt{3}) = 8 + 8(3) = 8 + 24 = \boxed{32}$$

Hence, option **(A)** is correct. **[1 mark]**

Q3. The roots of the quadratic equation $2x^2 - 5x + 3 = 0$ are: [1]

(A) $1, \frac{3}{2}$ **(B)** $-1, -\frac{3}{2}$

(C) $2, 3$ **(D)** $1, \frac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $1, \frac{3}{2}$

Splitting the middle term of $2x^2 - 5x + 3 = 0$ ($ac = 6$, $-2 - 3 = -5$):

Formula Used: $2x^2 - 2x - 3x + 3 = 0$

$$2x(x-1) - 3(x-1) = 0 \implies (x-1)(2x-3) = 0 \implies \boxed{x = 1, \frac{3}{2}}$$

Hence, option **(A)** is correct. **[1 mark]**

Q4. If p and q are the roots of the equation $x^2 + px + q = 0$, then: [1]

(A) $p = 1,\ q = -2$ **(B)** $p = 0,\ q = 1$

(C) $p = -2,\ q = 1$ **(D)** $p = -2,\ q = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $p = 1,\ q = -2$

Since p and q are the roots of $x^2 + px + q = 0$ ($a = 1, b = p, c = q$):

Formula Used: Sum $= p + q = -\frac{p}{1} = -p$, Product $= pq = \frac{q}{1} = q$

From the product relation $pq = q \implies q(p-1) = 0 \implies q = 0$ or $p = 1$.
Checking options: when $p = 1$, sum $1 + q = -1 \implies q = -2$ (Check: $x^2 + x - 2 = (x-1)(x+2) = 0$ has roots $1, -2 = p, q$).
(Also note if $q = 0 \implies p = 0$, which is not among options.) Thus, option **(A)** is correct. **[1 mark]**

Q5. If one root of the equation $ax^2 + bx + c = 0$ is three times the other, then $b^2 : ac =$: [1]

(A) $3 : 1$ **(B)** $3 : 16$

(C) $16 : 3$ **(D)** $16 : 1$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $16 : 3$

Let the roots of $ax^2 + bx + c = 0$ ($a \neq 0$) be α and 3α.

Formula Used: Sum: $\alpha + 3\alpha = -\frac{b}{a}$, Product: $\alpha(3\alpha) = \frac{c}{a}$

From the sum: $4\alpha = -\frac{b}{a} \implies \alpha = -\frac{b}{4a}$.
Substituting α into the product $3\alpha^2 = \frac{c}{a}$:

$$3\left(-\frac{b}{4a}\right)^2 = \frac{c}{a} \implies \frac{3b^2}{16a^2} = \frac{c}{a} \implies \frac{b^2}{ac} = \frac{16}{3} \implies \boxed{b^2 : ac = 16 : 3}$$

Hence, option **(C)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): If the coefficient of x^2 and the constant term have opposite signs, the equation $ax^2 + bx + c = 0$ always has real roots.

Reason (R): When a and c have opposite signs, $ac < 0 \implies -4ac > 0$, making $D = b^2 - 4ac > 0$ strictly positive.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Since a and c have opposite signs, $ac < 0$ (so $a \neq 0$, $c \neq 0$), which implies $-4ac > 0$.

Formula Used: $b^2 \geq 0$ and $-4ac > 0 \implies D = b^2 - 4ac > 0$

Because $D > 0$, the equation always has two real and distinct roots. Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Section B: Very Short Answer Questions

MATHS MANTHAN | $2 \times 2 = 4$ Marks

Q7. Determine whether the quadratic equation $2x^2 + 5\sqrt{3}x + 6 = 0$ has real roots, and if so, find them using the quadratic formula. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $2x^2 + 5\sqrt{3}x + 6 = 0$ with $ax^2 + bx + c = 0$, we get $a = 2$, $b = 5\sqrt{3}$, and $c = 6$.

Formula Used: $D = b^2 - 4ac$, $\quad x = \dfrac{-b \pm \sqrt{D}}{2a}$

Step 1: Compute discriminant D:

$$D = (5\sqrt{3})^2 - 4(2)(6) = 75 - 48 = 27 > 0 \quad \textbf{[1 mark]}$$

Since $D = 27 > 0$, the equation has **two distinct real roots**. Notice $\sqrt{D} = \sqrt{27} = 3\sqrt{3}$.

Step 2: Apply the quadratic formula:

$$x = \frac{-5\sqrt{3} \pm 3\sqrt{3}}{2(2)} = \frac{-5\sqrt{3} \pm 3\sqrt{3}}{4}$$

Taking $+$ sign: $x = \dfrac{-2\sqrt{3}}{4} = -\dfrac{\sqrt{3}}{2}$. Taking $-$ sign: $x = \dfrac{-8\sqrt{3}}{4} = -2\sqrt{3}$.

Hence, real roots exist and are $\boxed{x = -\dfrac{\sqrt{3}}{2} \text{ and } x = -2\sqrt{3}}$. **[1 mark]**

Q8. Find the values of k for which the quadratic equation $(k+4)x^2 + (k+1)x + 1 = 0$ has equal roots. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

For $(k+4)x^2 + (k+1)x + 1 = 0$ to be quadratic, $k + 4 \neq 0 \implies k \neq -4$.

Here $a = k + 4$, $b = k + 1$, and $c = 1$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

Setting $D = 0$:

$$(k+1)^2 - 4(k+4)(1) = 0$$
$$k^2 + 2k + 1 - 4k - 16 = 0 \implies k^2 - 2k - 15 = 0 \quad \textbf{[1 mark]}$$

Factorising ($-5 \times 3 = -15$ and $-5 + 3 = -2$):

$$k^2 - 5k + 3k - 15 = 0 \implies (k-5)(k+3) = 0 \implies k = 5 \text{ or } k = -3$$

Both 5 and -3 satisfy $k \neq -4$. Hence, $\boxed{k = 5 \text{ and } k = -3}$. **[1 mark]**

Q9. Solve the following equation for x: $\dfrac{x+1}{x-1} + \dfrac{x-2}{x+2} = 3, \quad x \neq 1, -2.$ **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Taking the common denominator $(x-1)(x+2)$ on the LHS ($x \neq 1, -2$):

Formula Used: $\dfrac{x+1}{x-1} + \dfrac{x-2}{x+2} = \dfrac{(x+1)(x+2) + (x-2)(x-1)}{(x-1)(x+2)}$

$$\frac{(x^2+3x+2) + (x^2-3x+2)}{x^2+x-2} = 3$$

$$\frac{2x^2+4}{x^2+x-2} = 3 \quad \textbf{[1 mark]}$$

Cross-multiplying:

$$2x^2 + 4 = 3(x^2+x-2) = 3x^2 + 3x - 6 \implies x^2 + 3x - 10 = 0 \quad \textbf{[1 mark]}$$

Factorising ($5 \times (-2) = -10$ and $5 - 2 = 3$):

$$x^2 + 5x - 2x - 10 = 0 \implies (x+5)(x-2) = 0 \implies x = -5 \text{ or } x = 2$$

Both values satisfy $x \neq 1, -2$. Hence, $\boxed{x = 2 \text{ and } x = -5}$. **[1 mark]**

Q10. A motorboat whose speed in still water is 18 km/h takes 1 hour more to travel 24 km upstream than to return downstream to the same spot. Determine the speed of the stream. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the speed of the stream be x km/h ($0 < x < 18$).
Upstream speed $= (18 - x)$ km/h; Downstream speed $= (18 + x)$ km/h. Distance $= 24$ km.

Formula Used: Time $= \dfrac{\text{Distance}}{\text{Speed}}$, $\quad t_{\text{upstream}} - t_{\text{downstream}} = 1$ hour

Setting up the equation:

$$\frac{24}{18-x} - \frac{24}{18+x} = 1$$

$$24\left[\frac{(18+x) - (18-x)}{324 - x^2}\right] = 1 \implies \frac{48x}{324-x^2} = 1 \implies x^2 + 48x - 324 = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Factorising ($54 \times (-6) = -324$ and $54 - 6 = 48$):

$$x^2 + 54x - 6x - 324 = 0 \implies x(x+54) - 6(x+54) = 0 \implies (x+54)(x-6) = 0 \quad \textbf{[1 mark]}$$

Thus $x = -54$ or $x = 6$. Since stream speed cannot be negative ($x > 0$), we reject $x = -54$. (Check: $\frac{24}{12} - \frac{24}{24} = 2 - 1 = 1$ h.)
Hence, the speed of the stream is $\boxed{6 \text{ km/h}}$. **[$\frac{1}{2}$ mark]**

Q11. A fast express train takes 3 hours less than a slow passenger train for a journey of 600 km. If the speed of the slow train is 10 km/h less than that of the fast train, formulate the quadratic equation and determine the average speeds of both trains. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the average speed of the fast express train be x km/h ($x > 10$).
Then the speed of the slow passenger train $= (x - 10)$ km/h. Journey distance $= 600$ km.

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}$, $\quad t_{\text{slow}} - t_{\text{fast}} = 3$ hours

Setting up the equation:

$$\frac{600}{x-10} - \frac{600}{x} = 3$$

$$600\left[\frac{x-(x-10)}{x(x-10)}\right] = 3 \implies \frac{6000}{x^2-10x} = 3 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

$$x^2 - 10x = 2000 \implies \boxed{x^2 - 10x - 2000 = 0} \quad \text{(required quadratic equation)} \quad \textbf{[1 mark]}$$

Factorising ($-50 \times 40 = -2000$ and $-50 + 40 = -10$):

$$x^2 - 50x + 40x - 2000 = 0 \implies x(x-50) + 40(x-50) = 0 \implies (x-50)(x+40) = 0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Thus $x = 50$ or $x = -40$. Since speed cannot be negative ($x > 10$), we reject $x = -40$.
So fast train speed $= 50$ km/h and slow train speed $= 50 - 10 = 40$ km/h. (Check: $\frac{600}{40} - \frac{600}{50} = 15 - 12 = 3$ h.)
Hence, $\boxed{\text{Fast express train} = 50 \text{ km/h}, \text{ Slow passenger train} = 40 \text{ km/h}}$. **[1 mark]**

Q12. If the roots of the quadratic equation $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$ are equal, prove that $a=b=c$. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Expanding each product in the given equation:

$$(x^2-(a+b)x+ab)+(x^2-(b+c)x+bc)+(x^2-(c+a)x+ca)=0$$
$$3x^2-2(a+b+c)x+(ab+bc+ca)=0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Comparing with standard form $Ax^2+Bx+C=0$, we have $A=3$, $B=-2(a+b+c)$, and $C=ab+bc+ca$.

Formula Used: Equal real roots $\iff D=B^2-4AC=0$

Setting $D=0$:

$$[-2(a+b+c)]^2-4(3)(ab+bc+ca)=0$$
$$4(a^2+b^2+c^2+2ab+2bc+2ca)-12(ab+bc+ca)=0 \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$
$$4(a^2+b^2+c^2-ab-bc-ca)=0$$

Dividing by 2 (leaving a factor of 2 inside to complete squares cleanly):

$$2(a^2+b^2+c^2-ab-bc-ca)=0$$
$$(a^2-2ab+b^2)+(b^2-2bc+c^2)+(c^2-2ca+a^2)=0$$
$$(a-b)^2+(b-c)^2+(c-a)^2=0 \quad \textbf{[1 mark]}$$

Since the sum of squares of real numbers can only be zero when each squared term is individually zero:

$$a-b=0,\quad b-c=0,\quad c-a=0 \implies \boxed{a=b=c} \qquad \text{(Hence Proved)} \quad \textbf{[1 mark]}$$

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Q13. Case Study: Urban Parks Landscaping & Triangular Garden Pavilion

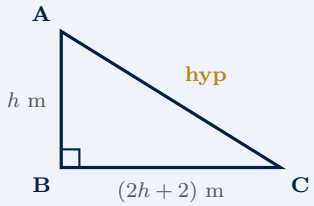
A municipal botanical park is constructing a triangular floral pavilion in the shape of a right-angled triangle. Environmental landscaping regulations specify that the horizontal base of the triangular pavilion must be 2 metres longer than twice its vertical altitude h metres. The total ground area enclosed by the floral pavilion is designed to be exactly 30 m^2. [4]

(i) Express the horizontal base of the triangular pavilion in terms of the vertical altitude h. **[1]**

(ii) Set up the quadratic equation in h representing the total enclosed area of the pavilion. **[1]**

(iii) Solve the equation to determine the altitude h, the base, and the hypotenuse of the pavilion. **OR** If ornamental perimeter kerbing costs Rs. 120 per metre, calculate the total cost of installing the decorative kerbing around all three sides of the triangular pavilion. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



As shown in the triangular pavilion layout alongside ($\triangle ABC$, $\angle B = 90°$), the vertical altitude is $AB = h$ m, the horizontal base is $BC = (2h + 2)$ m, and the enclosed area is 30 m^2.

We know the area and Pythagoras' formulas for a right-angled triangle:

Formula Used: Area $= \frac{1}{2} \times$ base $\times$ altitude, $\quad$ (Hypotenuse)^2 = (Base)^2 + (Altitude)^2

(i) Base Expression: Given horizontal base is 2 m longer than twice vertical altitude h m ($h > 0$):

$$\boxed{\text{Base } (b) = (2h + 2) \text{ m}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in h: Total enclosed ground area = 30 m^2:

$$\frac{1}{2} \times (2h + 2) \times h = 30 \implies (h + 1)h = 30 \implies \boxed{h^2 + h - 30 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Dimensions & Hypotenuse): Factorising $h^2 + h - 30 = 0$ ($6 \times (-5) = -30$):

$$(h + 6)(h - 5) = 0 \implies h = -6 \text{ or } h = 5$$

Since vertical altitude cannot be negative ($h > 0$), we reject $h = -6$. Thus $h = 5$ m, and Base $b = 2(5) + 2 = 12$ m.
By Pythagoras' theorem, Hypotenuse $= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ m.
Hence, $\boxed{\text{Altitude } h = 5 \text{ m, Base} = 12 \text{ m, Hypotenuse} = 13 \text{ m}}$. **[2 marks]**

OR (Alternative Part — Perimeter Kerbing Cost): Using the side lengths 5 m, 12 m, 13 m obtained in part (iii):

$$\text{Perimeter of triangular pavilion} = 5 + 12 + 13 = 30 \text{ m}$$

Cost of kerbing at Rs. 120 per metre $= 30 \times 120 =$ Rs. 3600.
Hence, the total kerbing cost is $\boxed{\text{Rs. } 3600}$. **[2 marks]**

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